Supplementary Table S4.1 of Chapter 4.

Comorbidities in Juvenile-onset Rheumatic Diseases: a systematic review and meta-analysis

Shall be considered in conjunction with the

Thesis submitted in accordance with the requirements of the

University of Liverpool

for the degree of Doctor of Philosophy by

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Supplementary Table S4.1. Prevalence and incidence rates of comorbidities in JIA, jSLE and JDM patients presenting in childhood or apparent in adulthood

Childhood / JIA										
Study	**Gender**	**Age in yrs**	**Onset age, yrs**	**Age diagnosis, yrs**	**Disease duration**	**Follow up period for the study**	**Comorbidity**	**Sample size, N (patients/ controls)**	**Prevalence**	**Incidence (per 1000 person-yrs)**	**Ethnicity /Race**
Demirkaya et al. (2011), Turkey[1]	F: 354	Mean 11.84±4.66 yrs			Mean 4.35 ±3.25 yrs	Mean 7.6±4.4 yrs		634			
							Macrophage Activation Syndrome	14	0.022082	2.905528806	
							Infections	14	0.022082	2.905528806	
							Uveitis	62	0.0977918	12.86734186	
Jain et al. (2001), India[2]	F: 10	Mean 8.7± 3.2 yrs			Mean 16.4±5.23 months			50			
							Keratoconjunctivitis sicca	7	0.14		
Rangel et al. (2003), UK[3]	F:20	Mean 15.2±2.4 yrs	Mean 5.5±3.6 yrs		Mean 9.8±3.8 yrs			30			Caucasian 27, Asian (SE) 3
							Major depressive episode	2/27	0.0740741		
							Over-anxious disorder/generalized anxiety disorder	3/27	0.1111111		
							Dysthymic disorder	2/27	0.0740741		
							Night terrors	1	0.0333333		
							Oppositional defiant disorder	1	0.0333333		

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Saurenmann et al. (2007), Canada[4]	F: 711	Mean 13.8 yrs (1.2-34.1)	Mean 6.9 yrs (0.5-17.8 yrs)	Mean 6.9 yrs (Range 0.08-25.7)	Mean 6.9 yrs (Range 0.08-25.7)		1081			
						Uveitis	142	0.1313599	19.03765971	
						Cataracts	33	0.0305273	4.424244862	
						Synechiae	31	0.0286772	4.15610881	
						Glaucoma	22	0.0203515	2.949496575	
						Band keratopathy	20	0.0185014	2.681360522	
						Macular oedema	7	0.0064755	0.938476183	
						Legal blindness	10	0.0092507	1.340680261	
						Impaired vision	6	0.0055504	0.804408157	
Chhabra et al. (2020), Canada[5]	F: 142	Median 16.9 yrs (IQR 12.9 -18.0)			Median 5.6 yrs (IQR 3.3-8.3)		247			
						Uveitis	24	0.097166	17.35106998	
						Band keratopathy	1	0.0040486	0.722961249	
						Decreased visual acuity	1	0.0040486	0.722961249	
						Cataract	1	0.0040486	0.722961249	
						Micrognathia or facial asymmetry	11	0.0445344	7.952573742	
						Leg-length discrepancy	12	0.048583	8.675534991	
Beukelman et al. (2012), USA[6]	F: 5,421	Mean 9.7±4.4 yrs			Median 1.1 yrs (IQR 0.5-2.2)		8479			White 4402, Black 1448, Latino 1721, other/unknown 932
						Upper respiratory tract	110	0.0129732	11.79384361	
						Pneumonia	87	0.0102606	9.327858131	
						Bacteraemia/septicaemia	67	0.0079019	7.183522928	
						Urinary tract/pyelonephritis	65	0.007666	6.969089408	
						Skin and soft tissue	44	0.0051893	4.717537445	

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						Abdominal abscess	20	0.0023588	2.144335202
						Gastroenteritis	30	0.0035382	3.216502804
						Diabetes mellitus	56	0.0066046	6.004138567
						Asthma	669	0.0789008	71.72801252
van Rossum et al. (2007), Netherlands[7]	F: 41	Median 18 yrs (range 10-25)		Median 10.7 yrs (range 8-23)	Median 9 yrs (IQR 8-9 yrs)		61		
						Uveitis	12	0.1967213	21.8579235
Gerloni et al. (2008), Italy[8]	F:124	Mean 17.1 yrs SD 9.2 yrs (range 1.9-49.8)	Mean 6.4 yrs SD 4.8 (Range 0.5-16.8)	Mean 10.7 yrs SD 8.6 (range 1.8-49.8)			163		
						Flare or new onset of chronic iridocyclitis	7	0.0429448	
						Gonadal effects	6	0.0368098	
						Hypertension	8	0.0490798	
						Infective events	41 (split below)		
						Recurrent URI or UTI or mycotic vaginitis	21	0.1288344	
						Viral thyroiditis	1	0.006135	
						Varicella	4	0.0245399	
						EBV infections	3	0.0184049	
						Single URI	3	0.0184049	
						Rotavirus gastro-enteritis	3	0.0184049	
						Helicobacter gastritis	1	0.006135	
						Scarlet fever	1	0.006135	
						Hip prosthetic infection	1	0.006135	
						CMV pneumonia complicated by MAS	1	0.006135	
						Pneumonia	1	0.006135	

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					Impetigo	1	0.006135
					Neuropsychiatric manifestations	53 (split below)	
					Headache	10	0.0613497
					Severe headache (dose-dependent)	4	0.0245399
					Psychoses	2	0.0122699
					Depression	3	0.0184049
					Panic attacks	2	0.0122699
					Anxiety	4	0.0245399
					Anorexia nervosa	1	0.006135
					Nervousness/hyperactivity	7	0.0429448
					Memory and concentration deficit	1	0.006135
					Aggressiveness	3	0.0184049
					Fatigue	5	0.0306748
					Hyperalgesia	1	0.006135
					Pain amplification syndrome	1	0.006135
					Sleepiness	2	0.0122699
					Insomnia	1	0.006135
					Vertigo	3	0.0184049
					Paraesthesia	1	0.006135
					Hypoglossal paralysis	1	0.006135
					Neuroalgodistrofy	1	0.006135
Klotsche et al. (2016), Germany[9]	F: 1,537	Mean 11.3 yrs (SD 4.6)	Mean 7.8 yrs (SD 3.7)	Mean 3.5 yrs (SD 3.6)		2263	
					Malignancies	8 (split below)	
					Acute lymphoblastic leukaemia	2	0.0008838

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				Thyroid gland carcinoma	1	0.0004419	
				Yolk sac carcinoma	1	0.0004419	
				Non-Hodgkin's lymphoma	1	0.0004419	
				Hodgkin's lymphoma	2	0.0008838	
				Oligodendroglioma	1	0.0004419	
				Uveitis	361	0.1595228	
				Inflammatory bowel disease (IBD)	14 (split below)		
				Colitis	4	0.0017676	
				Crohn's disease	10	0.0044189	
				Infections	471 (split below)		
				Bacterial	29	0.0128148	
				Viral	107 (split below)		
				Epstein Barr infection	8	0.0035351	
				Varicella zoster infection	34	0.0150243	
				Chickenpox	16	0.0070703	
				Herpes Zoster	18	0.007954	
				HPV	2	0.0008838	
Beukelman et al. (2013), USA[10]	F: 5,421	Median 10.1 yrs (IQR 5.9-13.3)	Median 1.2 yrs (IQR 0.6-2.4)		8,503		White 4,402, Black 1,448, Latino 1,721, Other/unknown 932
				Infections		(300/100,000 (95%CI 216-406) yrs split below)	
				Herpes zoster		225 /100,000 yrs (95%CI 154–318)	

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Salmonella	86/100,000 yrs (95%CI 44–150)
Coccidioides	35/100,000 yrs (95%CI 11–81)
Nontuberculous Mycobacterium	21/100,000 yrs (95%CI 4–61)
Nocardia	7/100,000 yrs (95%CI 0.2-39)
Pneumocystis	7/100,000 yrs (95%CI 0.2-39)
Toxoplasma	7/100,000 yrs (95%CI 0.2-39)
Blastomyces	0/100,000 yrs (95%CI 0.2-39)
Histoplasma	0/100,000 yrs (95%CI 0-36)
JC virus	0/100,000 yrs (95%CI 0-36)
Cryptococcus	0/100,000 yrs (95%CI 0-36)
Tuberculosis	0/100,000 yrs (95%CI 0-36)

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					Legionella			0/100,000 yrs (95%CI 0-36)	
					Listeria			0/100,000 yrs (95%CI 0-36)	
					Aspergillus			0/100,000 yrs (95%CI 0-36)	
Schenck et al. (2018), German[11]	F: 8,043	Mean 11.6 yrs		Mean 4.2 yrs		12,269			
					Type 1 Diabetes	58	0.0047274		
					Type 2 Diabetes	2	0.000163		
					Mature onset diabetes of the young	2	0.000163		
Angeles-Han et al. (2015), USA[12]	F: 208		Median 6.4 (2.9-11.7) yrs	Median 3.4, range (1.8 – 5.9) yrs	Mean 15.1± 12 months	287			Caucasian 221, African American 36, Others 30
					Uveitis	52	0.1811847	11.9989847	
					Cataracts	14	0.0487805	3.230495881	
					Glaucoma	7	0.0243902	1.615247941	
					Synechiae	12	0.0418118	2.76899647	
					Band keratopathy	11	0.0383275	2.538246764	
					Cystoid macular oedema	7	0.0243902	1.615247941	
					Cataract extraction	3	0.010453	0.692249117	

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Thiele et al. (2021), Germany[13]	F: 2,490	Median 12.3, IQR (8.6-15.2)				3,258			Caucasian 95%, African and Asian 5%
					Pneumonia	30	0.0092081		
					Herpes zoster	34	0.0104359		
					Varicella	23	0.0070595		
					SAE infection	108	0.0331492		
					Infections	813 (split below)			
					Upper respiratory tract infection	412	0.1264579		
					Otitis media	28	0.0085942		
					Urinary tract infection	32	0.009822		
					Tonsillitis	67	0.0205648		
					Gastroenteritis	63	0.019337		
					Streptococcal infection	20	0.0061387		
Kearsley-Fleet et al. (2022), multiple countries[14]	F: 2,014	Mean, 6.5 (SD 4.6) Median 14, IQR (10–17) yrs	Median 1, IQR (0-4) yrs, N=2,894	Median 5, IQR (3-9) yrs, N=2,926		2,693			
					MAS: Macrophage activation syndrome (systemic JIA only)	37/250	0.148	29.6	
					Varicella	822/2,238	0.3672922	73.45844504	
					Tuberculosis	17	0.0063127	1.262532492	
					Uveitis	556	0.2064612	41.29223914	
	F: 1,046	Mean 13.2 (SD 6.1) Median 22, IQR	Median 3, IQR (1-7) yrs, N=1,531	Median 14, IQR (7-18) yrs, N=1,514		1,541			

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		(19–25) yrs							
					Uveitis	238	0.1544452	11.03179753	
					MAS: Macrophage activation syndrome (systemic JIA only)	5/82	0.0609756	4.355400697	
					Varicella	127/871	0.1458094	10.41495818	
					Tuberculosis	2/1,541	0.0012979	0.092704181	
	F : 2,476	Mean 6.5 SD (4.5) Median 13, IQR (9–17) yrs	Median 3, IQR (1–6) yrs	Median 6, IQR (3–9) yrs		3,562			
					Uveitis	676/3,484	0.1940299	32.33830846	
					MAS: Macrophage activation syndrome (systemic JIA only)	62/366	0.1693989	28.23315118	
					Varicella	1,166/2,312	0.5043253	84.05420992	
					Tuberculosis	54/3,006	0.0179641	2.994011976	
Krause et al. (2017), USA[15]	F: 57	Mean 8.6 (±5.1) yrs at index	Median 1.3, range (0.2–4.0) yrs	Mean 6.3 (±4.4) yrs		89			Caucasian 75, Hispanic 2, Black 4, Asian 3, Other 1, Unknown 4
					Joint surgeries	6	0.0674157	10.70090958	
					Non-joint surgeries	14	0.1573034	24.96878901	
					Non-surgical hospitalisations (due to infection, JIA management etc.)	12	0.1348315	21.40181915	
					Hospitalisations for depression	5	0.0561798	8.917424648	
					Pregnancies	9/28	0.3214286	51.02040816	
					Depression	7	0.0786517	12.48439451	

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Lee et al. (2020), USA[16]	F: 10,233	Mean 13.86 ±3.44 yrs	Mean 2.68 yrs SD 2.34		15,210		
				Type 1 diabetes	18	0.0011834	44/100,000 person yrs (95%CI 28-70)
				Type 2 diabetes	27	0.0017751	66/100,000 person yrs (95%CI 45-96)
				Hyperlipidaemia	191.646	0.0126	4.701492537
				Hypertension	158.184	0.0104	3.880597015
				Obesity	305.721	0.0201	7.5
Horneff et al. (2022), Germany[17]		aged 2-15 yrs			546		
				Allergic rhinitis	76	0.1391941	
				Predominantly allergic asthma	15	0.0274725	
				Amyloidosis	<5	<5/546	
				Anaemia	15	0.0274725	
				Phobic anxiety disorders	17	0.0311355	
				Essential hypertension	<5	<5/546	
				Atopic dermatitis	80	0.1465201	
				Persistent somatoform pain disorder	46	0.0842491	
				Ulcerative colitis	<5	<5/546	
				Depression	21	0.0384615	
				Diabetes mellitus	5	0.0091575	
				Iron deficiency	8	0.014652	
				Post-viral fatigue syndrome	0	0	

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Fibromyalgia	13	0.0238095
Autoimmune thyroiditis	6	0.010989
Thyrotoxicosis	0	0
Hypothyroidism	8	0.014652
Lack of expected normal physiological development	26	0.047619
Crohn's disease	12	0.021978
Migraine	14	0.025641
Kidney disease (chronic)	<5	<5/546
Psoriasis	21	0.0384615
Sicca syndrome	5	0.0091575
Uveitis	59	0.1080586
	849	
Allergic rhinitis	94	0.1107185
Predominantly allergic asthma	25	0.0294464
Amyloidosis	11	0.0129564
Anaemia	24	0.0282686
Phobic anxiety disorders	30	0.0353357
Essential (primary) hypertension	<5	<5/849
Atopic dermatitis	105	0.1236749
Persistent somatoform pain disorder	71	0.0836278
Ulcerative colitis	<5	<5/849
Depression	19	0.0223793
Diabetes mellitus	7	0.008245
Iron deficiency	7	0.008245
Post-viral fatigue syndrome	<5	<5/849

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					Fibromyalgia	9	0.0106007
					Autoimmune thyroiditis	13	0.0153121
					Thyrotoxicosis	<5	<5/849
					Hypothyroidism	18	0.0212014
					Lack of expected normal physiological development	31	0.0365135
					Crohn's disease	<5	<5/849
					Migraine	29	0.0341578
					Kidney disease (chronic)	0	0
					Osteoporosis	5	0.0058893
					Psoriasis	39	0.0459364
					Sicca syndrome	<5	<5/849
					Uveitis	90	0.1060071
Bolt et al. (2008), Switzerland[18]	F: 174	≤18 yrs	Mean 6.49, range (0.67–15.67) yrs	Mean 4.82, range (0.08–16.58) yrs		265	
					Synechiae	10	0.0377358	7.829014327
					Cataracts	9	0.0339623	7.046112894
					Band keratopathy	5	0.0188679	3.914507164
					Glaucoma	3	0.0113208	2.348704298
					Macular oedema	2	0.0075472	1.565802865
					Uveitis	35	0.1320755	27.40155014
					Impaired visual acuity	1	0.0037736	0.782901433
					Legal blindness in one eye	2	0.0075472	1.565802865
Simon et al. (2020), USA[19]		< 18 yrs					

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	F: 6,173	Mean 10.6±4.5 yrs				8,633		
					Multiple sclerosis	1	0.0001158	
					Asthma	777	0.0001158	
					Anxiety	295	0.0001158	
					Depression	208	0.0001158	
					Diabetes mellitus	92	0.0001158	
					Anaemia	498	0.0001158	
					Hypertension	2	0.0001158	
					Hyperlipidaemia	174	0.0001158	
					Malignancy	56	0.0001158	
	F: 6,587	Mean 10.5±4.5 yrs				9,155		
					Multiple sclerosis	1	0.0001092	
					Asthma	887	0.0001092	
					Anxiety	317	0.0001092	
					Depression	214	0.0001092	
					Diabetes mellitus	84	0.0001092	
					Anaemia	479	0.0001092	
					Hypertension	6	0.0001092	
					Hyperlipidaemia	202	0.0001092	
					Malignancy	49	0.0001092	
Schulz et al. (2022), Germany[20]	F 333	Mean 6.8 yrs (range 2-22)	Median 6.0 yrs (<1-16.4)	Median 4.4 yrs (range 0.9–14.1)		499		Europe 413, Asia 31, Turkish 28, Africa Arab 2, mixed countries of origin 30 and
					Hypothyroidism	6	0.012024	
					Autoimmune thyroiditis	8	0.0160321	
					Other thyroid disorder	2	0.004008	
					Coeliac disease	3	0.006012	
					Scleroderma	1	0.002004	

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						Overlap syndrome	1	0.002004	unknown 23
Swart et al. (2018), Italy[21]	F: 5,584	≤ 18 yrs	Median 5.4 yrs (IQR 2.4–10.0)	Med 6.2 (IQR 2.8–10.9) yrs	Med 5.3 (IQR 2.7–8.8) yrs		8,274		
						Infections	1,523 (split below)		
						Serious/targeted infections (Epstein–Barr virus, cytomegalovirus, papilloma virus, herpes zoster primary and reactivation, and opportunistic infections)	674	0.08146	
						Tuberculosis	27	0.0032632	
						Other infections	822	0.0993474	
						Macrophage activation syndrome	75	0.0090645	
						Autoimmune disease	50 (split below)		
						Inflammatory bowel disease (IBD)	21	0.0025381	
						Multiple sclerosis	2	0.0002417	
						Other autoimmune diseases excluding IBD, uveitis, and demyelinisation disorder	18	0.0021755	
						Malignancies	16 (split below)		
						Leukaemia's	3	0.0003626	

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			Lymphomas	2	0.0002417
			Hematopoietic neoplasms (excluding leukaemia and lymphoma)	1	0.0001209
			Neoplasm (other)	10	0.0012086
			Gastrointestinal (GI) ulcer/GI bleed/GI perforation	17	0.0020546
			Pregnancy	9	0.0010877
			Congestive heart failure	1	0.0001209
F: 2,670	Median 7.6 yrs (IQR 3.2–11.7)	Med 6.1 (IQR 3.5–9.5) yrs		3,990	
			Infections	1,509 (split below)	
			Serious/targeted infections (Epstein–Barr virus, cytomegalovirus, papilloma virus, herpes zoster primary and reactivation, and opportunistic infections)	171	0.0428571
			Tuberculosis	0	0
			Other infections	1,338	0.3353383
			Macrophage activation syndrome	11	0.0027569
			Autoimmune disease	50 (split below)	
			Inflammatory bowel disease (IBD)	23	0.0057644

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				Multiple sclerosis	0	0
				Other autoimmune diseases excluding IBD, uveitis, and demyelinisation disorder	24	0.006015
				Malignancies	13 (split below)	
				Leukaemia	2	0.0005013
				Lymphoma	5	0.0012531
				Hematopoietic neoplasms (excluding leukaemia and lymphoma)	2	0.0005013
				Neoplasm (other)	4	0.0010025
				Gastrointestinal (GI) ulcer/GI bleed/GI perforation	4	0.0010025
				Pregnancy	7	0.0017544
				Congestive heart failure	0	0
Giancane et al. (2019), Italy[22]	F: 189	Med 3.1 yrs (IQR 1.8–5.8)	Med 7.7 yrs (IQR 6.1–10.2)		239	
				Ocular	21	0.0878661
				Muscular atrophy	19	0.0794979
				Osteoporosis	3	0.0125523
				Subcutaneous atrophy	9	0.0376569
				Growth failure	19	0.0794979
				Pubertal delay	5	0.0209205
				Avascular necrosis	0	0
				Amyloidosis	0	0
				Diabetes	0	0

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	F: 219	Med 2.5 yrs (IQR 1.6–4.7)	Med 6.9 yrs (IQR 5.4–9.7)		269		
				Ocular	7	0.0260223	
				Muscular atrophy	8	0.0297398	
				Osteoporosis	1	0.0037175	
				Avascular necrosis	1	0.0037175	
				Subcutaneous atrophy	6	0.0223048	
				Growth failure	1	0.0037175	
				Diabetes	0	0	
				Amyloidosis	0	0	
Grassi et al. (2007), Italy[23]	F: 244	mean 4.9 ± 3.6 yrs	mean 7.6 ± 5.6 yrs		309		
				Synechiae	12	0.038835	5.109862034
				Band keratopathy	8	0.02589	3.406574689
				Cataract	6	0.0194175	2.554931017
				Glaucoma	2	0.0064725	0.851643672
				Retinal detachment	1	0.0032362	0.425821836
				Iris atrophy	1	0.0032362	0.425821836
				Blindness	4	0.012945	1.703287345
				Uveitis	62	0.2006472	26.40095384
				Severe visual loss	8	0.02589	3.406574689
Horneff et al. (2016), Germany[24]			13,198 observation yrs		3,695		
				Suspected Malignancies	12 (split below)		
				Hodgkin's lymphoma	2	0.0005413	4.10117E-05
				non-Hodgkin's lymphoma	1	0.0002706	2.05058E-05
				acute lymphatic leukaemia (ALL)	2	0.0005413	4.10117E-05

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				lymphoproliferative disorder	1	0.0002706	2.05058E-05
				thyroid carcinoma	1	0.0002706	2.05058E-05
				yolk sac carcinoma	1	0.0002706	2.05058E-05
				anaplastic ependymoma	1	0.0002706	2.05058E-05
Lovell et al. (2021), USA[25]	F: 867	Mean 6.6 yrs (SD 4.5)	Mean 8.0 (SD 4.8)		1,332		
				Autoimmune diseases	1,085 (split below)		
				Addison disease	11	0.0082583	
				Autoimmune/Hashimoto's thyroiditis	16	0.012012	
				Graves' disease	1	0.0007508	
				Morphoea	12	0.009009	
				Multiple sclerosis	1	0.0007508	
				Takayasu arteritis	1	0.0007508	
				Type 1 diabetes mellitus	17	0.0127628	
				Vitiligo	5	0.0037538	
				Myasthenia Gravis	0	0	
				Pernicious Anaemia	0	0	
				Primary biliary cirrhosis	0	0	
				Associated conditions		0	
				Autoimmune disease not elsewhere classified	8	0.006006	
				Chronic glomerulonephritis	1	0.0007508	
				Chronic urticaria	49	0.0367868	
				Episcleritis/scleritis	5	0.0037538	
				Erythema nodosum	1	0.0007508	

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					Haemolytic anaemia	1	0.0007508	
					Immune thrombocytopenia purpura	2	0.0015015	
					Leukocytoclastic vasculitis	2	0.0015015	
					Pulmonary fibrosis/interstitial lung disease	15	0.0112613	
					Raynaud's syndrome	19	0.0142643	
					Sjögren's syndrome/sicca syndrome	3	0.0022523	
					Thrombocytopenia purpura	1	0.0007508	
					Vasculitis	8	0.006006	
Barthel et al. (2015), Germany[26]	F: 2,073	Mean 6.1 ± 3.9 yrs		8,389 patient-yrs		3,071		
					Inflammatory bowel disease	11 (split below)		
					Crohn Disease	8	0.002605	0.000310527
					Ulcerative colitis	3	0.0009769	0.000116448
Sahin et al. (2021), Turkey[27]	F: 289	8.8 ± 4.7 yrs (mean) (range 9 months to 17 yrs)	4.0 ± 1.9 yrs (mean)			500		
					Uveitis	34	0.068	
					Osteoporosis	8	0.016	
					Delayed puberty	6	0.012	
					Autoimmune Thyroiditis	4	0.008	

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				Inflammatory Bowel Disease (IBD)	2	0.004	
				Depression	2	0.004	
				Hypertension	1	0.002	
				Amyloidosis	1	0.002	
				Familial mediterranean fever (FMF)	63	0.126	
				Psoriasis	15	0.03	
				Cataract	4	0.008	
				Band keratopathy	1	0.002	
				Posterior synechia	1	0.002	
Pohjankoski et al. (2010), Finland[28]	F: 283		Mean 7 (1–15) yrs		417		
				Uveitis	91	0.2182254	
				Diabetes mellitus type 1	9	0.0215827	
				Coeliac disease	3	0.0071942	
				Hypothyroidism	3	0.0071942	
Cakan et al. (2020), Turkey[29]	F: 202		Mean 29.2±18.0 months		405		
				Uveitis	27	0.0666667	2.283105023
Cosickic et al. (2017), Bosnia and Herzegovina [30]	F: 61	Med 7.1 yrs (IQR 2.3 -10.6 yrs)			97		
				Synechia	3	0.0309278	
				Band keratopathy	2	0.0206186	
				Cataracts	2	0.0206186	
				Glaucoma	1	0.0103093	
				Uveitis	14	0.1443299	

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Berthold et al. (2022), Sweden[31]	F: 433			Med 8.7 yrs ((IQR 3.7-12.7)		640	
					Depressive disorder	93	0.1453125
					Anxiety disorder	111	0.1734375
Lien et al. (2003), Norway[32]	F: 80	Mean 17.0±1.8 yrs	Mean 2.8±1.4 yrs	Mean 14.2±1.4 yrs		105	
					Osteopenia/Low total-body BMC	42/103	0.407767
					Osteoporosis (BMD Z score ≤ -2.5)	5/103	0.0485437
Saraux et al. (2015), France[33]	F: 18	Mean 13.5 yrs (SD 5.84)				26	
					Infections	3	0.1153846
					Bacterial infection	3	0.1153846
					Viral infection	1	0.0384615
Alberdi-Saugstrup et al. (2017), Nordic countries[34]	F: 135		Median 6.0 (IQR 2.8-10.0)			193	
					Uveitis	30	0.1554404
Amine et al. (2011), Morrocco[35]	F: 32	Mean 11 ± SD 3.3 yrs range (2-16)	Mean 7±SD 3.4 range (1–15)	Mean 3 ± SD 2.4 yrs range (1-12)		58	
					Obesity	13	0.2241379
Adulthood / JIA							

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Smitherman et al. (2023), USA[36]	F: 144	Mean 20±1.4 yrs	CARRA (or any other) JIA registry, Age onset will have childhood onset disease.				187		White 147, Hispanic 23, Black 9, Other 8
						Self-reported symptoms of pain or discomfort	122	0.6524064	
						Self-reported symptoms of anxiety or depression	84	0.4491979	
Mena-Vazquez et al. (2022), Spain[37]	F: 35	Mean 25±6.1 yrs		161.4 months (103.6 - 220.4 IQR)	24 months		52		Caucasian 50
						Depression	13	0.25	125
						Arterial hypertension	1	0.0192308	9.615384615
						Diabetes melllitus	1	0.0192308	9.615384615
						Dyslipidaemia	4	0.0769231	38.46153846
						Obesity (BMI ≥30)	7	0.1346154	67.30769231
						Uveitis	11	0.2115385	105.7692308
Raab et al. (2013), Germany[38]	F: 196	Mean 19.7 yrs (SD 2.8)		Mean 10.6 yrs (since onset) SD 5.6			344		
						Cardiovascular diseases	34 (split below)		
						Hypertension	25	0.0726744	6.856077227
						Cardiac insufficiency	1	0.002907	0.274243089

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Pulmonary diseases	28 (split below)		
Asthma	26	0.0755814	7.130320316
Gastrointestinal diseases	14 (split below)		
Crohn's disease	4	0.0116279	1.096972356
Ulcerative colitis	3	0.0087209	0.822729267
Gastrointestinal ulcer	1	0.002907	0.274243089
Liver and gallbladder diseases	3	0.0087209	0.822729267
Diseases of the kidneys or the Urinary tract	16 (split below)		
Renal insufficiency	1	0.002907	0.274243089
Kidney stone	4	0.0116279	1.096972356
Amyloidosis	3	0.0087209	0.822729267
Gynaecological diseases	12	0.0348837	3.290917069
Haematological diseases	23 (split below)		
Anaemia	11	0.0319767	3.01667398
Iron deficiency	6	0.0174419	1.645458534
Allergic diseases	84 (split below)		
Allergic rhinitis	50	0.1453488	13.71215445
House dust mite allergy	16	0.0465116	4.387889425
Thyroid/endocrine diseases	23 (split below)		
Hypothyroidism	7	0.0203488	1.919701624
Hashimoto thyroiditis	4	0.0116279	1.096972356
Diabetes mellitus	3	0.0087209	0.822729267
Skin diseases	67 (split below)		

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					Psoriasis	22	0.0639535	6.03334796
					Atopic dermatitis	30	0.0872093	8.227292672
					Eye diseases	80 (split below)		
					Uveitis	59	0.1715116	16.18034226
					Cataract	3	0.0087209	0.822729267
					Glaucoma	9	0.0261628	2.468187802
					Other eye diseases	8 (split below)		
					Benign eye tumour	1	0.002907	0.274243089
					Episcleritis	1	0.002907	0.274243089
					Scleritis	2	0.005814	0.548486178
					Sicca syndrome/ sicca symptoms	3	0.0087209	0.822729267
					Chronic conjunctivitis	1	0.002907	0.274243089
					Astigmatism	1	0.002907	0.274243089
					Neurologic diseases	41 (split below)		
					Migraine	30	0.0872093	8.227292672
					Epilepsy	5	0.0145349	1.371215445
					Psychiatric disorders	32 (split below)		
					Depression	17	0.0494186	4.662132514
					Anxiety disorder	7	0.0203488	1.919701624
					Other comorbidities	30 (split below)		
					Osteoporosis	11	0.0319767	3.01667398
Aggarwal et al. (2006), India[39]	M: 30	Mean 22.6 ± 5.12 yrs	Mean 11.37 ± 2.67 yrs	Mean 11.27 ± 5.39 yrs		30		
					Osteopenia (Lumbar Spine)	14	0.4666667	41.40786749
					Osteopenia (Hip)	15	0.5	44.36557232

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						Osteopenia (Distal third of forearm)	15	0.5	44.36557232	
						Osteoporosis (Lumbar Spine)	8	0.2666667	23.66163857	
						Osteoporosis (Hip)	7	0.2333333	20.70393375	
						Osteoporosis (Distal third of forearm)	12	0.4	35.49245785	
						Joint deformities	5	0.1666667	14.78852411	
Arvidsson et al. (2010), Norway[40]	F: 44	Mean 35.2±3.6 yrs (27.9-44.4)	Mean 5.5±3.1 yrs (0.8-14.5)	Mean 29.7±3.3 yrs (24.4-38.0)	Mean 27 yrs		60			
						Craniofacial growth disturbance	34	0.5666667	20.98765432	
						Micrognathia	16	0.2666667	9.87654321	
Packham et al. (2002), UK[41]	F: 176	Mean 35.4±11.1 yrs (range 19-78)	Mean 7.1±4.5 yrs (range 0.8–15.9)	Mean 28.3±10.8 yrs (range 8–73)			246			Asian 1, Afro-Caribbean 1, Caucasian descent 244
						Uveitis	22%	0.22	7.77385159	
						Glaucoma	4.10%	0.041	1.448763251	
						Amyloidosis	8.90%	0.089	3.144876325	
						Cataracts	10.20%	0.102	3.604240283	
						Required eye surgery	10.60%	0.106	3.745583039	
						Sicca syndrome	15.00%	0.15	5.300353357	
						At least one major prosthetic joint replacement	51.2% (split below)			
						Hip prosthetic	43.90%	0.439	15.51236749	
						Revision hip prostheses	15.40%	0.154	5.441696113	
						Knee prosthetic	27.60%	0.276	9.752650177	
						Shoulder prosthetic	4.90%	0.049	1.731448763	
						Elbow prosthetic	3.30%	0.033	1.166077739	
						Wrist prosthetic	1.60%	0.016	0.565371025	

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						Ankle prosthetic	1.60%	0.016	0.565371025
Mars et al. (2019), Finland[42]	F: 92	Mean 32.4 ± 13.4 yrs					119		
						Cardiovascular diseases	17	0.1428571	
						Diabetes	5	0.0420168	
						Eye disorders	16	0.1344538	
						Gastrointestinal disorders	11	0.092437	
						Infections	27	0.2268908	
						Malignancies	2	0.0168067	
						Mental disorders	12	0.1008403	
						Neurological disorders	18	0.1512605	
						Other musculoskeletal disorders	38	0.3193277	
						Rheumatic diseases	76	0.6386555	
						Skin disorders	19	0.1596639	
Dimopoulou et al. (2017), Greece[43]	F: 72	Mean 25±5.9, median 24, (range 8-41) yrs	Median 7, Mean 7.7±4 (range 1-15.5) yrs	Median 17 (range 5-31.5) yrs	Mean 17.2±6.7 yrs		102		
						Uveitis	11	0.1078431	6.26994984
						Prosthetic arthroplasty	9	0.0882353	5.12995896
						Mandibular correction	1	0.0098039	0.56999544
						Arthrodesis of the radiocarpal joint	1	0.0098039	0.56999544
						Hands or foot small joint correction	4	0.0392157	2.27998176
						Blindness in one eye	2	0.0196078	1.13999088
						Leg length discrepancy ≥0.5cm	37	0.3627451	21.08983128
						Mandibular dysfunction (such as reduced mouth	29.40%	0.0028824	0.167578659

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					opening, frontal facial asymmetry or retrognathia)				
Rebane et al. (2022), Finland[44]	F: 62	Mean 23 yrs (range 16-23)		17 yrs		97			
					Coeliac disease	1	0.0103093	0.606428138	
					Crohn's disease	2	0.0206186	1.212856277	
					Irritable bowel syndrome (IBS)	2	0.0206186	1.212856277	
					Hypothyroidism	1	0.0103093	0.606428138	
					Depression	2	0.0206186	1.212856277	
					Panic disorder	2	0.0206186	1.212856277	
					Uveitis	4	0.0412371	2.425712553	
					Ulcerative colitis	0	0	0	
Minden et al. (2019), Germany[45]	F: 464	Mean 23.4±3.9 yrs	Mean 14.3±6.1 yrs	Mean 9.1±3.7 yrs		701			
					Arthroplasty	36	0.0513552	5.643429324	
					Eye surgery	22	0.0313837	3.448762365	
					Uveitis	63	0.0898716	9.876001317	
					Synovectomy	112	0.1597718	17.55733567	
Smith et al. (2019), USA[46]	F: 170	Mean 30.55±0.39 yrs (estimated)	Mean 20.32±7.32 yrs			170			White 145, African American 3, Asian/Pacific islander 2, Native America 3, Other 1
					Comorbid pregestational conditions				
					Thyroid dysfunction	13	0.0764706		
					Hypertension	21	0.1235294		
					Diabetes Mellitus (Type 1 and 2)	0	0		
					Other autoimmune diseases				

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					Inflammatory bowel disease	11	0.0647059
					Sjögren's syndrome or UCTD	3	0.0176471
					Antiphospholipid syndrome	1	0.0058824
					Psoriasis	11	0.0647059
					Lupus	4	0.0235294
					Obstetric complications		
					Preeclampsia	15	0.0882353
					PIH without preeclampsia	10	0.0588235
					Placenta previa and/or abruption	4	0.0235294
					Preterm labour	25	0.1470588
					GDM	13	0.0764706
					Fever during pregnancy	47	0.2764706
					Caesarean section	70	0.4117647
Barth et al. (2016), Germany[47]	F: 1,617	Range (18–73) yrs	Median 8, range (0–15), IQR (4-11) yrs	Median 27, range (3–70), IQR (15-40) yrs		2,592	
					Pain/discomfort	(split below)	
					Moderate problems	1,395	0.5381944
					Severe problems	94	0.0362654
					Anxiety/depression:	(split below)	
					Moderate problems	624	0.2407407
					Severe problems	54	0.0208333
					Severe disability	783	0.3020833

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Simon et al. (2020), USA[19]		≥ 18 yrs			1 year			
	F: 4,506	Mean 37.0 ±16.5 yrs				5,789 (split below)		
					Multiple sclerosis	20	0.0034548	3.454828122
					Asthma	472	0.0815339	81.53394369
					Anxiety	555	0.0958715	95.87148039
					Depression	653	0.1128001	112.8001382
					Diabetes mellitus	440	0.0760062	76.00621869
					Anaemia	948	0.1637589	163.758853
					Hypertension	103	0.0177924	17.79236483
					Hyperlipidaemia	1,162	0.2007255	200.7255139
					Malignancy	278	0.0480221	48.0221109
	F: 4,315	Mean 34.6 ±14.7 yrs				5,638 (split below)		
					Multiple sclerosis	19	0.00337	3.369989358
					Asthma	486	0.0862008	86.20078042
					Anxiety	549	0.097375	97.37495566
					Depression	622	0.1103228	110.3228095
					Diabetes mellitus	371	0.0658035	65.80347641
					Anaemia	655	0.1161759	116.1759489
					Hypertension	103	0.0182689	18.26888968
					Hyperlipidaemia	1,023	0.1814473	181.4473217
					Malignancy	162	0.0287336	28.73359347
Packham and Hall, (2003), UK[48]	F: 177	Mean 35.4±11.1 yrs (19-78)	Mean 7.1±4.2 yrs (range 0.8-15.9)	Mean 28.7±10.8 yrs (range 8-73)		177		
					Primary ovarian failure	9 (split below)		
					Primary ovarian failure (unrelated to medication use)	6	0.0338983	1.181125613

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Ma et al. (2022), Taiwan[49]	F: 1,155	0-30 yrs	249,593 months		2,791		
				Obstructive sleep apnoea (OSA)	31	0.0111[49]071	0.64 per 10000 person yrs 95%CI 0.5-0.82)
				Allergic rhinitis	401	0.1436761	0.006907699
				Obesity	19	0.0068076	0.000327297
				Osteoporosis	7	0.0025081	0.000120583
				Disorders of mineral metabolism	3	0.0010749	5.16785E-05
				Hypocalcaemia/hypomagnesemia	0	0	0
Zhang-Jian et al. (2020), Taiwan[50]	F: 549	Age at pregnancy is 24.4 ± 3.4 yrs			549		
				Pregnancy outcomes			
				Stillbirth	4	0.007286	
				Foetal distress	32	0.0582878	
				Foetal abnormalities, any	38	0.0692168	
				Preeclampsia	17	0.0309654	
				Gestational diabetes	24	0.0437158	
				Pregnancy-related hypertension	24	0.0437158	
				Antepartum haemorrhage	55	0.1001821	
				Severe postpartum haemorrhage	25	0.0455373	
				Chorioamnionitis	5	0.0091075	
				Caesarean delivery	155	0.2823315	

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					Low birth weight <2500g	64	0.1165756	
					Prematurity <37 weeks	49	0.0892532	
Rypdal et al. (2021), USA[51]	F: 297	≥19 yrs	Mean 5.7 IQR 2.6-9.8	17.6 yrs IQR (16.7-18.4)		434		
					Uveitis	96	0.2211982	12.56807708
					At least 1 ocular complication	31/80	0.3875	22.01704545
					Cataract	25/80	0.3125	17.75568182
					Glaucoma	22/80	0.275	15.625
					Synechiae	14/80	0.175	9.943181818
					Macular oedema	8/80	0.1	5.681818182
					Band keratopathy	7/80	0.0875	4.971590909
					Epiretinal membrane	3/80	0.0375	2.130681818
					Hypotony	3/80	0.0375	2.130681818
					Phthisis	3/80	0.0375	2.130681818
					Blindness <3/60 - binocular	2/80	0.025	1.420454545
					Blindness <3/60 - monocular	3/80	0.0375	2.130681818
Childhood / jSLE								
Davis et al. (2023), USA[52]	F: 571	Mean 18.4±3.7 yrs (10-24)				650		White 367, Black 110, Hispanic 91, Asian 41, Unknown 41
					Depression	117	0.18	

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					Anxiety	78	0.12	
					Adjustment disorder	65	0.1	
					Seizure/stroke disorder	52	0.08	
Garf et al, 2015, Egypt[53]	F: 49	Mean 13.1±3.0 yrs		Mean 3.9±3.0 yrs (range 0.2-12 yrs)		70		
					Vitamin D deficiency	28	0.4	
					Vitamin D insufficiency (≤30/ng/ml)	42	0.6	
Paim-Marques et al. 2019, Brazil[54]	F: 61	Mean 17.9±3.07 yrs	Mean 13.56 ± 2.60 yrs (corneal verticillata) 12.63 ± 2.9 (no corneal verticillata)		Mean 5.06 ± 2.04 yrs (with corneal verticillata)	76		
					Visual acuity reduction	25	0.3289474	65.00936135
					Posterior subcapsular cataracts	4	0.0526316	10.40149782
					Keratitis	40	0.5263158	104.0149782
					Crystalline abnormalities	1	0.0131579	2.600374454
					Corneal vortex keratopathy	16	0.2105263	41.60599126

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Silva et al. 2015, Brazil[55]	F: 732	Median 14.5 (6–20) yrs, IFI, 17.0 (2–25.9) non- IFI	Median 12.4 (range 5.1–17.2) yrs, IFI, 11.8 (0.2–17.8) non IFI		Median 1.0 (range 0–7.3) yrs, IFI, 4.7 (range 0–23.4) non- IFI		852	White 592/830
						Invasive Fungal Infections (Proven)	22 (split as follows)	
						Candidiasis	13	0.0152582
						Aspergillosis	6	0.0070423
						Cryptococcosis	2	0.0023474
						Histoplasmosis	1	0.0011737
						Arterial hypertension	119/843	0.1411625
						Acute renal failure	37/841	0.0439952
						Chronic renal failure	27/842	0.0320665
						Renal replacement Therapy	28/842	0.0332542
Knight et al. 2016, USA[56]	F: 805	Mean 14.7 SD 2.0 (10-18 yrs)					970	White 141, African American 410, Latino 265, Other 154
						Depression diagnosis	181	0.1865979
						Anxiety	72	0.0742268
						Acute stress/adjustment	60	0.0618557
						Other psychiatric disorder	175	0.1804124
						Seizures/stroke	155	0.1597938
Das Chagas Medeiros et al. 2016, Brazil[57]	F: 52	Mean 15.5±2.0 yrs			Mean 10.4±6.4 yrs		60	

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			Antiphospholipid syndrome	3/49	0.0612245	5.886970173
			Hypothyroidism	1/37	0.027027	2.598752599
			Thrombosis	7/60 (split below)		
			Deep vein thrombosis	1	0.0166667	1.602564103
			Mesenteric artery thrombosis	1	0.0166667	1.602564103
			CVA	1	0.0166667	1.602564103
			Pulmonary embolus	1	0.0166667	1.602564103
			Portal vein thrombosis	1	0.0166667	1.602564103
			Renal vein thrombosis	1	0.0166667	1.602564103
			DVT + PE	1	0.0166667	1.602564103
			Neoplasia	1/58	0.0172414	1.657824934
			hypertension	27/60	0.45	43.26923077
			Cardiovascular diseases	3/58 (split below)		
			Congestive heart failure			
			Coronary artery disease			
			Aortic and mitral valve disorder			
			Diabetes	0/60	0	0
			Seizure	7/60	0.1166667	11.21794872
			Psychosis	2/60	0.0333333	3.205128205
Frittoli et al. 2016, Brazil[58]	F: 38	Mean 14.5±2.84 yrs		41		
			Cognitive dysfunction	17	0.4146341	
			Anxiety symptoms	8	0.195122	
			Depression symptoms	1	0.0243902	

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Holland et al. 2018, USA[59]	F: 869	Mean 12.2±3.1 yrs		Mean 3.81±2.98 yrs	Mean 3.8 yrs		1048	
						Ocular		
						Cataracts	4.20%	4.008E-05	0.010546404
						Retinal change	1.70%	1.622E-05	0.004268783
						Neuropsychiatric			
						Cognitive impairment	8.40%	8.015E-05	0.021092808
						Seizures requiring therapy	3.90%	3.721E-05	0.00979309
						Cerebrovascular accident	2.9%/0.4%	0.0069179	1.820510245
						Cranial/peripheral neuropathy	2.40%	2.29E-05	0.006026517
						Transverse myelitis	0.40%	3.817E-06	0.001004419
						Renal			
						End-stage renal disease	1.30%	1.24E-05	0.003264363
						Pulmonary			
						Pulmonary hypertension	0.40%	3.817E-06	0.001004419
						Pleural fibrosis	1.30%	1.24E-05	0.003264363
						Shrinking lung	0.80%	7.634E-06	0.002008839
						Pulmonary fibrosis	0.90%	8.588E-06	0.002259944
						Pulmonary infarction	0%	0	0
						Cardiovascular			
						Angina/artery bypass	0.10%	9.542E-07	0.000251105
						Myocardial infarction	0.1 / 0.1	0.0009542	0.251104861
						Cardiomyopathy	0.90%	8.588E-06	0.002259944
						Valvular disease	0.90%	8.588E-06	0.002259944
						Pericarditis/pericardiectomy	1.60%	1.527E-05	0.004017678
						Peripheral vascular			

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Claudication for 6 months	0.30%	2.863E-06	0.000753315
minor tissue loss	2.40%	2.29E-05	0.006026517
significant tissue loss	0.7/0.3%	0.0022265	0.585911343
Venous thrombus swelling, ulcer/stasis	2.50%	2.385E-05	0.006277622
Gastrointestinal			
Bowel infarction or resection	1.3 / 0.3%	0.0041349	1.088121066
Mesenteric insufficiency	0.10%	9.542E-07	0.000251105
Peritonitis	0.20%	1.908E-06	0.00050221
Stricture/upper GI tract surgery ever	0.40%	3.817E-06	0.001004419
Pancreatic insufficiency	0.50%	4.771E-06	0.001255524
Musculoskeletal			
Muscle atrophy/weakness	7.90%	7.538E-05	0.019837284
Deforming/erosive arthritis	4.70%	4.485E-05	0.011801928
Osteoporosis with fracture	2.60%	2.481E-05	0.006528726
Avascular necrosis	2.7 / 1.0	0.0025763	0.677983126
Osteomyelitis	0.40%	3.817E-06	0.001004419
Ruptured tendons	0.10%	9.542E-07	0.000251105
Skin			
Scarring chronic alopecia	9.10%	8.683E-05	0.022850542
Extensive scar/panniculum	1.70%	1.622E-05	0.004268783
Skin ulceration for 6 months	2.70%	2.576E-05	0.006779831
Premature gonadal failure	2%	1.908E-05	0.005022097

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						Diabetes	0.90%	8.588E-06	0.002259944	
						Malignancy	0.20%	1.908E-06	0.00050221	
Gad et al. 2013, Egypt[60]	F: 17	Mean 12.90±2.67 3 yrs (8-16 yrs)		Mean 4.40±1.875 yrs			20			
						Sensorineural hearing loss (SNHL)	5	0.25		
						Mild Conductive hearing loss (CHL)	2	0.1		
Jongvilaikasem and Rianthavorn, 2021, Thailand[61]	F: 87	Mean 11.3 ± 2.4 yrs			26 months		97			
						Short final height	22	0.2268041	104.6788263	
						Mild CKD	88	0.9072165	418.7153053	
						Advanced CKD	9	0.0927835	42.82315623	
AlAhmed et al. 2020, USA[62]	F: 306	Mean 11.39±3.8 yrs	Mean 12.04±3.62 year (Age first seen by paediatric rheumatologist)				386			Hispanic 102, Non-Hispanic 283, White 216, African American 105, Asia 33, Other 13
						Cerebrovascular accident	2	0.0051813		
						Cranial nerve disease	1	0.0025907		
						Renal dialysis	3	0.007772		
						Arterial thrombosis	3	0.007772		
						Venous thrombosis	10	0.0259067		
						Avascular necrosis	5	0.0129534		
						Autoimmune thyroiditis	10	0.0259067		
						Coeliac disease	1	0.0025907		
						T1DM	1	0.0025907		
						Other autoimmune disorder	22	0.0569948		

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			Pulmonary hypertension	0	0	
			Coronary heart disease	0	0	
			Renal transplant	0	0	
Karve, S. et al, 2012, USA[63]	0-15 yrs	18 months		71		
			Asthma	8	0.1126761	75.11737089
			Atopic dermatitis	2	0.028169	18.77934272
			Allergic rhinitis	6	0.084507	56.33802817
			General diabetes	1	0.0140845	9.389671362
			Rheumatoid arthritis	14	0.1971831	131.4553991
			Multiple sclerosis	1	0.0140845	9.389671362
			Thyroiditis	1	0.0140845	9.389671362
			Fracture in post-index period	4	0.056338	37.55868545
			Psoriasis	0	0	0
			Type 1 diabetes	0	0	0

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					Other CNS demyelinating disease	0	0	0
					Vitiligo	0	0	0
					Grave's disease	0	0	0
Ye, Q. et al. 2021, China[64]	F: 82	Median age 169.02 months (range 31.00-220.53)	133.83, range (27.70–202.83) months	28.00, range (0–144.00) months		107		
					Cataracts	21	0.1962617	
					Steroid-induced osteonecrosis of femoral head	3	0.0280374	
					Osteoporosis	5	0.046729	
					Short stature	8	0.0747664	
					Obesity	7	0.0654206	
					Diabetes	6	0.0560748	
					HDL resistance	1	0.0093458	
					Upper respiratory tract infection	11	0.1028037	
					Pneumonia	13	0.1214953	
					EB virus infection	15	0.1401869	
					Progressive systemic sclerosis	1	0.0093458	
					Secondary Sjogren	6	0.0560748	
					Juvenile rheumatoid arthritis	1	0.0093458	
					Antiphospholipid syndrome	2	0.0186916	

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Cann et al. 2022, Australia[65]	F: 37		12.5 yrs (mean) 13 yrs (median)	Mean 4.3 yrs , IQR (2.5–6.1) yrs		42			Caucasian 16, Indigenous Australian 11, Asian 10, Other 5
					Cataract	2	0.047619	11.07419712	
					retinal change	2	0.047619	11.07419712	
					CVA	3	0.0714286	16.61129568	
					Significant tissue loss	1	0.0238095	5.53709856	
					Venous thrombosis	1	0.0238095	5.53709856	
					Muscle weakness	1	0.0238095	5.53709856	
					AVN	1	0.0238095	5.53709856	
					Scarring alopecia	2	0.047619	11.07419712	
					Diabetes	1	0.0238095	5.53709856	
					Growth failure	3	0.0714286	16.61129568	
					Delayed puberty	1	0.0238095	5.53709856	
					Premature gonadal failure	0	0	0	
					Malignancy	0	0	0	
Medhat et al. 2020, Egypt[66]	F: 86	Mean 12.7 ±2.6 yrs (range 6–16)	Mean 9.9±2.5, range (4–15) yrs	Median 24, IQR (12–48) months		106			
					Osteoporosis	6	0.0566038		
					Hypothyroidism	2	0.0188679		
					Diabetes mellitus	4	0.0377358		
					Hypertension	31	0.2924528		
					Hepatitis C virus infection	5	0.0471698		
					Chronic kidney disease	5	0.0471698		
					End-stage renal disease	1	0.009434		
					Avascular necrosis	1	0.009434		
					Sicca manifestations	2	0.0188679		

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						Ischaemic heart disease	0	0
						CVA	0	0
						Malignancy	0	0
Aggarwal et al. 2018, India[67]	F: 250	≤ 18 yrs	Median 14 yrs range (1-17)		Median 3.5 yrs, Range (0.25-28 yrs) (248 patients), 25 patients were not followed up		273	
						Cushingoid habitus	35	0.1282051	36.63003663
						Growth retardation	32	0.1172161	33.4903192
						Avascular necrosis	9	0.032967	9.419152276
						Cataract	8	0.029304	8.372579801
						Osteopenia	10	0.03663	10.46572475
						Infection	72	0.2637363	75.35321821
						Hypertension	42	0.1538462	43.95604396
						DVT	8	0.029304	8.372579801
						Pulmonary thromboembolism	2	0.007326	2.09314495
						Stroke	3	0.010989	3.139717425
						Renal vein thrombosis	1	0.003663	1.046572475
Al-Mayouf et al. 2008, Saudi Arabia[68]	F: 76	F 14.4±3.9 yrs, male 15.8±4.4 yrs (mean)	Mean 9.3±2.8 yrs(F), 10±2.5 yrs (M)	F 66.5 ± 39.2 months, Male 76.2 ± 56.4 months, median 5 yrs	F 60.5 ± 38 months, Male 69.1 ± 57.5 months		89	
						Diabetes	1	0.011236	0.185718265
						Malignancy	0	0	0
						Dialysis + transplant (ESRD)	5	0.0561798	0.928591327
						Premature gonadal failure	0	0	0
						Transverse myelitis	2	0.0224719	0.371436531

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				Pyoderma gangrenosum	1	0.011236	0.185718265	
				Pancreatic pseudocyst	1	0.011236	0.185718265	
Costagliola et al. 2018, Italy[69]	F: 24	Mean 14.6 ± 1.6 yrs	Mean 14.17 ± 8.04 yrs (1 patient excluded)		25			Caucasian 24, Hispanic 1
				Renal damage (including nephrotic syndrome or ESRD)	9	0.36	25.40578687	
				Vascular involvement (including DVT or ischaemic arterial lesions)	7	0.28	19.76005646	
				Nervous system involvement (with ischemic stroke, cranial neuropathies and cerebral vasculitis)	4	0.16	11.29146083	
				Cutaneous damage (chronic non-scarring alopecia)	3	0.12	8.468595625	
				Myositis	1	0.04	2.822865208	
				Acute myocardial infarction	1	0.04	2.822865208	
				Pulmonary (restrictive syndrome and pulmonary heart)	1	0.04	2.822865208	
				CMV reactivation	3	0.12	8.468595625	
				Herpes zoster reactivation	2	0.08	5.645730416	
				Severe EBV infection	1	0.04	2.822865208	
				Mycobacterium tuberculosis infection	1	0.04	2.822865208	

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					ESRD	3	0.12	8.468595625	
					Requiring dialysis	1	0.04	2.822865208	
					Multi-infarct encephalopathy	1	0.04	2.822865208	
Torrente-Segarra et al. 2017, Spain[70]	F: 435	Mean 31.5±10.5 Median 30 range (24-28) yrs Childhood	Mean 16.6 ± 6.3 yrs, median 16 (13-18)	Mean 10.7±7.2 yrs		484			Caucasian 441
					Articular				
					Erosive arthritis	40	0.0826446		
					Atrophy, muscle weakness	15	0.0309917		
					Tendon rupture	5	0.0103306		
					Osteoporosis	15	0.0309917		
					Fibromyalgia	11	0.0227273		
					Avascular necrosis	1.6 ± 0.62 (1-2)			
					Pulmonary embolism	11	0.0227273		
					Lung fibrosis	6	0.0123967		
					Pulmonary hypertension	8	0.0165289		
					Alveolar haemorrhage	6	0.0123967		
					Shrinking lung	4	0.0082645		
					Pleural fibrosis	1	0.0020661		
					Pulmonary infarction	1	0.0020661		
					Vascular disease	25	0.0516529		
					Valvular dysfunction	19	0.0392562		
					Libman-Sachs Endocarditis	8	0.0165289		
					Cardiomyopathy	7	0.0144628		

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Angina or cardiac bypass	6	0.0123967
Myocarditis	3	0.0061983
Heart attacks	1.1± 0.01 (1-1)	
Small tissue loss	22	0.0454545
Major tissue loss	3	0.0061983
Venous thrombosis	3	0.0061983
Intermittent claudication >6 months	1	0.0020661
Renal terminal failure	27	0.0557851
Seizures (requiring therapy for 6 months)	46	0.0950413
Cognitive impairment/ Psychosis	21	0.0433884
Neuropathy	13	0.0268595
Transverse myelitis	3	0.0061983
Strokes	1.2±0.61 (1-1)	
Cataract	31	0.0640496
Retinopathy	22	0.0454545
Amenorrhoea <40	22 (51.2%)	0.0454545
Dyslipidaemia	104	0.214876
High blood pressure	100	0.2066116
Autoimmune thyroiditis	31	0.0640496
Alcoholism	14	0.0289256
Congestive heart failure	8	0.0165289
Diabetes mellitus with organ involvement	8	0.0165289

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					Diabetes mellitus without organ involvement	0	0		
					Chronic obstructive pulmonary disease	7	0.0144628		
					Peripheral arterial disease	3	0.0061983		
					Malignancy Solid without metastasis	15	0.0309917		
					Malignancy Solid with metastasis	0	0		
					Leukaemia	1	0.0020661		
					Lymphoma	2	0.0041322		
					Dialysis	23	0.0475207		
					Kidney transplantation	15	0.0309917		
					Splenectomy	14	0.0289256		
Ponin et al. 2022, Thailand[71]	F: 87	Mean 20.6 ± 3.0 yrs	Mean 12.1 ± 2.3 yrs	Mean 5.5 ± 2.7 yrs		106			
					Growth impairment	25	0.2358491		
Avar Aydin et al. 2018, USA[72]	F: 87		Mean yrs 13.2±3.1		Median 29.5 months (IQR 6.0–55.2)	110			Caucasian 54, African American 39, Asian 8, Other 9
					Hypertension	22	0.2	81.3559322	
					Obesity	19	0.1727273	70.26194145	
					Dyslipidaemia	14	0.1272727	51.77195686	
Ahluwalia et al. 2014, India[73]	F: 17				Mean 82 months (range 9–210 months)	21 (27 patients initially, 6 lost to follow up)			

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					Thrombosis	3 (split below)		
					Cerebral infarct (death)	1	0.047619	6.968641115
					Pulmonary thromboembolism (death)	1	0.047619	6.968641115
					Spinal infarct	1	0.047619	6.968641115
Sinicato et al. 2017, Brazil[74]	F: 37	16.7 ± 4.0 (mean)		Mean 4.8 ± 4.1 yrs		41		
					Iglyceridemia	11	0.268	
					Insulin resistance	2	0.049	
					Arterial hypertension	4	0.097	
					Albumin Exretion	5	0.122	
					High LDL-c levels	19	0.463	
					Hypercholesterolemia	8	0.195	
Cabral et al. 2013, Portugal[75]	F: 46		Mean 12.6±4.04 yrs Median 13.5 range (1-17)	Mean 5.5±5.4 yrs Median 4 range (0-24)		56		
					Cataract	4	0.0714286	12.98701299
					Epilepsy	3	0.0535714	9.74025974
					Obesity	25	0.4464286	81.16883117
					Cardiac failure	2	0.0357143	6.493506494
					Arterial hypertension	5	0.0892857	16.23376623
					Avascular necrosis	1	0.0178571	3.246753247
					Growth delay	3	0.0535714	9.74025974
					Osteoporosis	5	0.0892857	16.23376623
					Gastrointestinal tract bleeding	1	0.0178571	3.246753247

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				Macrophage activation syndrome	2	0.0357143	6.493506494
				Infection	6 (split below)		
				Pneumonia	2	0.0357143	6.493506494
				Sepsis	1	0.0178571	3.246753247
				Bacterial endocarditis	1	0.0178571	3.246753247
				Cellulitis	3	0.0535714	9.74025974
				Cognitive impairment	4	0.0714286	12.98701299
				End stage renal disease	2	0.0357143	6.493506494
				Vascular thrombosis	5 (split below)		
				Cerebrovascular (thrombosis)	4	0.0714286	12.98701299
				Cerebral hemorrhage	2	0.0357143	6.493506494
				Myocardium infarct	1	0.0178571	3.246753247
				Epilepsy	3	0.0535714	9.74025974
Bernatsky et al. 2017, North America[76]	F: 978	Mean 13±3.3 yrs	Mean 7.6 yrs, SD 6.4 yrs, IQ 3.4-10.0 yrs/8839 PY		1168		
				Non-Hodgkin lymphoma	3	0.0025685	0.337959625
				Head and neck squamous cell carcinomas (tongue and nasopharynx)	2	0.0017123	0.225306417
				Glioblastoma	1	0.0008562	0.112653208
				Thyroid cancer (papillary adenoma)	1	0.0008562	0.112653208
				Breast cancer (infiltrating duct carcinoma)	1	0.0008562	0.112653208

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					Adenocarcinoma not otherwise specified	1	0.0008562	0.112653208	
					Cancers whose morphology and histology codes were indeterminate	3	0.0025685	0.337959625	
Yang et al. 2015, Canada[77]	F: 30 (AVN), F: 82 (non-AVN)		Mean 13.8 ± 2.5 (AVN), 13.8 ± 2.4 (non-AVN)	Mean 6.89 ± 5.39 yrs (AVN), 5.72 ± 5.09 yrs (non-AVN)		617			AVN: South Asian 10, Other Asian 9, Black 9, White 8, Other 1. Non- AVN: South Asian 10, Other Asian 30, Black 12, WhIte 40, Hispanic 1 Other 13, Unknown 5
					Symptomatic Avascular Necrosis (AVN)	37	0.0599676	8.703568228	
Lin et al. 2018, Taiwa[78]			Mean 12.9±3.12 yrs (range 6-18 yrs)	mean 6 yrs		35			
					Vitamin D deficiency	28	0.8	133.3333333	
Kahwage et al. 2017, Brazil[79]	F: 6/7 (uveitis)	≤25 yrs, Median 10.4 range (8.8 -14.5) (Uveitis)				852			

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					Uveitis	7	0.008216
					Cataracts & bilateral irreversible blindness	1	0.0011737
					Unilateral blindness	1	0.0011737
Lai et al. 2022, China[80]	F: 95	Age at enrolment - Girls mean 14.4 (SD 3.4), Boys mean 13.9 (SD 5.0)		Girls mean 6.6 (SD 4.1) Boys mean 5.2 (SD 4.0)		115	
					Hypertension	22	0.1913043
					Chronic kidney disease	14	0.1217391
Adulthood / jSLE							
Al Hamzi et al. 2014, Saudi Arabia[81]	F: 45	Mean 23.6±4 yrs	Mean 10.3±6 yrs	Mean 15±4 yrs		48	
					Growth retardation/Short stature	24	0.5
					Delayed puberty	13	0.2708333
					Renal disease	12 (as split below)	
					Progressive renal disease and required dialysis	7	0.1458333
					Renal transplant	5	0.1041667
					Neuropsychiatric	9	0.1875
					Cardiovascular	6	0.125
					Musculoskeletal	9	0.1875
					Skin	2	0.0416667

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					Seizure disorder	6	0.125	
					Chorea	3	0.0625	
					Cerebrovascular accident	3	0.0625	
					Miscarriage	4	0.0833333	
Tanigava et al. 2022, Brazil[82]	F: 65	Median 21 yrs (IQR 20-26)	Median 14 IQR 12-16	Median 9 yrs IQR 6-14		69		White 35, Black 31, Asian 3
					Mental disorders	16	0.2318841	
					Obesity	14	0.2028986	
					Thyroid diseases	7	0.1014493	
					Fibromyalgia	6	0.0869565	
					Antiphospholipid syndrome	56	0.8115942	
					Osteoporosis and osteoporotic fracture	5	0.0724638	
					Hypertension	5	0.0724638	
					Cerebrovascular disease	4	0.057971	
					Allergic disorders	4	0.057971	
					Gastrointestinal diseases	4	0.057971	
					Valvular disease	4	0.057971	
					Chronic kidney disease	3	0.0434783	
					Diabetes mellitus	2	0.0289855	
					Dyslipidaemia	2	0.0289855	
					Retinal change or optic atrophy	5	0.0724638	
					Cataract	2	0.0289855	
					Cognitive impairment or major psychosis	3	0.0434783	

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Seizures requiring therapy for ≥6 months	2	0.0289855
Cerebrovascular accident	3	0.0434783
Cranial or peripheral neuropathy	3	0.0434783
Transverse myelitis	0	0
End-stage renal disease	3	0.0434783
Pulmonary hypertension	0	0
Pulmonary fibrosis	2	0.0289855
Shrinking lung	0	0
Pleural fibrosis	0	0
Pulmonary infarction	0	0
Angina or coronary artery bypass	0	0
Myocardial infarction	0	0
Cardiomyopathy	0	0
Pericarditis or pericardiectomy	0	0
Venous thrombosis	1	0.0144928
Infarction or resection of bowel	1	0.0144928
Mesenteric insufficiency	0	0
Chronic peritonitis	0	0
Stricture or upper gastrointestinal tract surgery	0	0
Pancreatic insufficiency	1	0.0144928
Muscle atrophy or weakness	0	0

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					Deforming or erosive arthritis	0	0
					Osteoporosis with fracture or vertebral collapse	2	0.0289855
					Avascular necrosis	2	0.0289855
					Osteomyelitis	0	0
					Ruptured tendon	0	0
					Scarring chronic alopecia	0	0
					Extensive scarring	1	0.0144928
					Skin ulceration (excluding thrombosis)	4	0.057971
					Premature gonadal failure	0	0
					Diabetes requiring therapy	0	0
					Malignancy	0	0
Hersh et al.2009, USA[83]	F: 78	Mean 30.5±12.8 yrs	14.0 ±2.7 range (2–17)	Mean 16.5 yrs ± 8.5 (range 3-47)		90	White 48.9%, Asian/Pacific Islander 16.7%, Hispanic/Latino 15.6 %, African American 7.8%, Other 11%
					Pulmonary	23.60%	0.236
					Clotting disorders	11.30%	0.113
					Seizures	23.60%	0.236
					Myocardial infarction	7.90%	0.079
					Transplant	12.4	0.1377778
					Dialysis	19.1	0.1377778
Sousa et al. 2016, Portugal[84]	F: 77	Mean 25.1 ± 9.1 yrs	13.4 ± 5.5			89	Caucasian 78, African 3, Others 4
					Hypertension	17	0.1910112
					Diabetes	3	0.0337079
					Hypercholesterolemia	2	0.0224719

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						Thyroid disease	3	0.0337079		
						Antiphospholipid syndrome	10	0.1123596		
						Sjögren's syndrome	2	0.0224719		
Koutsonikoli et al. 2015, Greece[85]	F: 39	Mean 22±7.8 yrs at the end of follow up	Mean 12.5 ± 3. 1 (SD)	Median 7.4 yrs and IQR (3.9–14.2) yrs	440 person-yrs (avg follow up time)		47			All Caucasians
						Cataract	4	0.0851064	0.193423598	
						Retinopathy	1	0.0212766	0.048355899	
						Optic atrophy	1	0.0212766	0.048355899	
						Cataract and retinopathy	1	0.0212766	0.048355899	
						Seizures	2	0.0425532	0.096711799	
						Cognitive impairment and seizures	1	0.0212766	0.048355899	
						Major psychosis and seizures	1	0.0212766	0.048355899	
						Cerebral vascular accident	1	0.0212766	0.048355899	
						Transverse myelitis	1	0.0212766	0.048355899	
						Single venous thrombosis	5	0.106383	0.241779497	
						Significant tissue loss and venous thrombosis	1	0.0212766	0.048355899	
						Avascular necrosis	3	0.0638298	0.145067698	
						Persistent impairment of renal function or end-stage renal disease	0	0	0	
						Pulmonary fibrosis	1	0.0212766	0.048355899	
						Pulmonary infarction	1	0.0212766	0.048355899	
						Premature gonadal failure	3	0.0638298	0.145067698	

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						Delayed growth	6	0.1276596	0.290135397
Gamal et al. 2022, Egypt[86]	F: 163	25 yrs (median), range, 20 to 30.3 yrs	14 (median) (range 12-15)		12 yrs (median), (range, 6.75 to 17.25) yrs		186		
						Hypertension	82	0.4408602	
						Diabetes mellitus	12	0.0645161	
						Dyslipidaemia	75	0.4032258	
						Thyroid dysfunction	10	0.0537634	
						Malignancy	1	0.0053763	
						Osteoporosis	27/150	0.18	
						Avascular necrosis	23	0.1236559	
						End stage renal disease	17	0.0913978	
						Thrombosis	22	0.1182796	
						Raynaud's phenomenon	51	0.2741935	
						Pulmonary hypertension	24	0.1290323	
Groot et al. 2019, Netherlands[87]	F: 101	Median 33 yrs (range 18-65)	14 (range 4-17)		Median 20 yrs range (1-55 yrs)		111		White 80, Non-White 31
						Muscle atrophy or weakness	1	0.009009	
						Deforming or erosive arthritis	10	0.0900901	
						Osteoporosis with fracture	6	0.0540541	
						Avascular necrosis	14	0.1261261	
						Osteomyelitis	1	0.009009	
						Ruptured tendon	1	0.009009	

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Cognitive impairment or psychosis	12	0.1081081
Seizures	10	0.0900901
CVA	7	0.0630631
Cranial or Peripheral Neuropathy	4	0.036036
Transverse Myelitis	0	0
ESRD	8	0.0720721
Claudication	0	0
Minor tissue loss	0	0
Significant tissue loss	4	0.036036
Venous thrombosis	8	0.0720721
Scarring alopecia	8	0.0720721
Scarring/ panniculum	0	0
Skin ulceration	3	0.027027
Cataracts	8	0.0720721
Retinal change	3	0.027027
Angina/bypass	4	0.036036
MI	5	0.045045
Cardiomyopathy	1	0.009009
Valvular disease	2	0.018018
Pericarditis	0	0
PAH	1	0.009009
Pulmonary fibrosis	4	0.036036
Shrinking lung	1	0.009009
Pleural Fibrosis	0	0
Pulmonary Infarction	0	0
Infarction or resection of bowel	4	0.036036
Mesenteric insufficiency	0	0
Chronic peritonitis	0	0

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					Stricture or upper GI surgery	0	0	
					Pancreatitis	0	0	
					Premature gonadal failure	2	0.018018	
					Malignancy	2	0.018018	
					IDDM	1	0.009009	
Sinicato et al. 2017, Brazil[74]	F:32					35		
					Low HDL-cholesterol	5	0.143	
					Hypertriglyceridemia	6	0.171	
					Insulin Resistance	0	0.00	
					Arterial Hypertension	4	0.14	
					Albumin excretion	4	0.114	
Childhood / JDM								
Huber et al. 2000, Canada[88]	F: 46	Median 13 (range 7-26)	Median 5.8 yrs Range (1-16)	Median 7.2 yrs range (3.2 - 13.9 yrs)		65		Caucasian 56, Asian 7, African Canadian 1, Indian 1
					Calcinosis	22	0.3384615	47.00854701
					Insulin-dependent diabetes mellitus	3	0.0461538	6.41025641
					Connective tissue diseases	2 (split below)		
					Scleroderma	1	0.0153846	2.136752137
					Overlap connective tissue disease	1	0.0153846	2.136752137
					Malignancies	0	0	
Cho et al. 2019, Korea[89]	F: 68	Mean 12.7 ± 7.1 yrs		10 yrs		130		

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Congestive heart failure	1	0.0076923	0.769230769
Cardiac arrhythmias	1	0.0076923	0.769230769
Valvular disease	1	0.0076923	0.769230769
Pulmonary circulation disorders	2	0.0153846	1.538461538
Peripheral vascular disorders	3	0.0230769	2.307692308
Hypertension, uncomplicated	5	0.0384615	3.846153846
Hypertension, complicated	2	0.0153846	1.538461538
Paralysis	3	0.0230769	2.307692308
Other neurological disorders	2	0.0153846	1.538461538
Chronic pulmonary disease	36	0.2769231	27.69230769
Diabetes, uncomplicated	8	0.0615385	6.153846154
Diabetes, complicated	2	0.0153846	1.538461538
Renal failure	1	0.0076923	0.769230769
Liver disease	15	0.1153846	11.53846154
Peptic ulcer disease excluding bleeding	15	0.1153846	11.53846154
AIDS/HIV	1	0.0076923	0.769230769
Solid tumour without metastasis	1	0.0076923	0.769230769
Coagulopathy	4	0.0307692	3.076923077
Weight loss	1	0.0076923	0.769230769
Fluid and electrolyte disorders	8	0.0615385	6.153846154
Deficiency anaemia	17	0.1307692	13.07692308
Depression	6	0.0461538	4.615384615
Pulmonary disease specified			

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							Interstitial lung disease	2	0.0153846	1.538461538	
Concannon and Han, 2021, New Zealand[90]	F: 18	≤15 yrs		Mean 8.6 yrs SD 3.7 yrs		Mean 6.5, range (0.4–16.7) yrs		31			Asian 5, Maori 7, pacific island 3, European 16
							Calcinosis	9	0.2903226	44.66501241	
							Cutaneous vasculopathy	5	0.1612903	24.81389578	
							Interstitial lung disease	6	0.1935484	29.77667494	
							Pneumomediastinum	2	0.0645161	9.925558313	
							Pneumothorax	3	0.0967742	14.88833747	
							Surgical emphysema	1	0.0322581	4.962779156	
Okong'o et al. 2016, South Africa[91]	F: 16		Median 6.75, range (2.0–9.7) yrs	Median 7.9 , range (3.4 – 9.75) year	median 4, range (0.5-84) months	Minimum of 12 months, median 55.5 months (inactive disease) 49.1 months (active disease)		25			Indigenous African 13, mixed race 11, European ancestry 1
							Skin				
							Calcinosis	8	0.32		
							Skin ulcers	1	0.04		
							Oedema (Generalized, Periorbital or limb)	0	0		
							Alopecia	3	0.12		
							Lipodystrophy	1	0.04		
							Musculoskeletal:		0		
							Arthritis	2	0.08		
							Contractures	5	0.2		
							Osteoporosis with fracture	1	0.04		
							Endocrine:		0		
							Growth failure	8	0.32		

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				Diabetes mellitus	1	0.04
				Adrenal insufficiency	0	0
				Respiratory:		0
				Interstitial lung disease (ILD)	2	0.08
				Pulmonary haemorrhage	1	0.04
				Gastrointestinal:		0
				Abdominal pain or bleeding	0	0
				Ocular:		0
				Cataracts	2	0.08
				Nervous system:		0
				Sensorineural hearing loss	1	0.04
				Seizures, neuropathy		0
				Cardiovascular		0
				Raynaud's phenomenon	6	0.24
				Cardiomyopathy	0	0
				Infections:		0
				Fungal (skin)	7	0.28
				Bacterial (staph aureus 2, gram negative 1)	3	0.12
				TB	2	0.08
Ravelli, A. et al, 2010, Italy, UK[92]	F: 321	Mean 6.9, range (0.9–17.8) yrs	Mean 7.7, range (2–25.2) yrs		490	White 320
				Cutaneous scarring/atrophy	195/443	0.4401806
				Calcinosis	106/406	0.2610837
				Joint contractures	82/448	0.1830357
				Lipodystrophy	43/444	0.0968468

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					Hirsutism	44/431	0.1020882	
					Growth failure	34/424	0.0801887	
					Osteoporosis with fractures	26/435	0.0597701	
					Cataract	9/430	0.0209302	
					Gastrointestinal infarction/resection	7/444	0.0157658	
					Malignancy	0/447	0	
Tabarki et al. 1998, France[93]	F: 22	Mean 6.9 yrs (range 2-14 yrs)		Mean 4.9 yrs (range: 2.5-13 yrs)		36		
					Calcinosis	15	0.4166667	85.03401361
					Impaired function	8	0.2222222	45.35147392
					Cataracts	1	0.0277778	5.66893424
					Vertebral collapse	2	0.0555556	11.33786848
					Contractures	1	0.0277778	5.66893424
Chevalier et al. 2021, France[94]	F: 4	Median 6 yrs range 2-11 yrs	Median 5.5 months	Mean 14.9±8.8 yrs		7			Caucasian 100%
					Polycystic ovary syndrome	1	0.1428571	9.587727709
					Depressive disorder	1	0.1428571	9.587727709
					mild physical dysfunction (CHAQ/HAQ >0)	2	0.2857143	19.17545542
					Calcinosis	2	0.2857143	19.17545542
					Lipodystrophy	2	0.2857143	19.17545542
					Osteopenia/osteoporosis	0	0	0
					Cardiovascular damage	0	0	0
					Peripheral vascular damage	0	0	0

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					Ocular damage	0	0	0
					Malignancy	0	0	0
					Fracture	0	0	0
					Moderate or severe disability	0	0	0
					CHAQ/HAQ>1			
Al-Mayouf et al. 2017, Saudi Arabia[95]	F: 58	Mean 6 ± 3 yrs		Mean 5 ± 4.4 yrs		92		
					Permanent skin changes	16	0.173913	34.7826087
					Residual muscle weakness	12	0.1304348	26.08695652
					Calcinosis	27	0.2934783	58.69565217
					Growth failure	37	0.4021739	80.43478261
					Osteoporosis	15	0.1630435	32.60869565
					Lipodystrophy	3	0.0326087	6.52173913
					Mucinous cystic ovarian tumour	1	0.0108696	2.173913043
Sag et al. 2021, Turkey[96]	F: 35	Mean 8.1±4.3 yrs	Mean 8.7±4.4	Median 4.85 yrs IQR 3.7, Mean 5.66±3.59 yrs		58		
					Calcinosis	19	0.3275862	67.54354781
					Pulmonary involvement	8	0.137931	28.43938855
					Skin ulcers	4	0.0689655	14.21969428
					Gastrointestinal involvement	4	0.0689655	14.21969428
					Severe muscle disease (wheel-chair dependency)	3	0.0517241	10.66477071
					Cardiac involvement	2	0.0344828	7.109847138

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Ramanan et al. 2005, Canada[97]								
	F: 20		Mean 8.4 yrs	Mean 34.6 months	JDM patients after June 1997	31		Black 2, East Indian 4, Native Canadian 1, Asian 1, White 23
					Calcinosis	2	0.0645161	22.37553608
					Cataracts	3	0.0967742	33.56330412
	F:16		6.7 yrs (mean)	4 yrs (1056 patient months)	JDM patients pre-June 1997	22		Black 1, East Indian 1, Native Canadian 1, Asian 2, White 17
					Calcinosis	5	0.2272727	56.81818182
					Cataracts	8	0.3636364	90.90909091
Sun et al. 2015, Taiwan[98]	F: 26	Mean 81.97 ± 46.63 months (25-216)	Mean 87.97± 45.90 months (28-213)	For those receiving IVIG: Mean 79 months		39		
					Calcinosis	15	0.3846154	
					Pulmonary involvement	4	0.1025641	Follow up time and cases were not reported coherently. Hence, no incidence rate was calculated
					Malignancy	0	0	
					Disability	4	0.1025641	

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					GI perforation	1	0.025641	
					Avascular necrosis of the hip joint	1	0.025641	
Silverberg et al. (2018), USA[99]	F: 915	0-17				1407		White 543, Black 203, Hispanic 260, Asian 19, Multiracial /other 71
					Obesity	38	0.0270078	
					Hypertension	119	0.0845771	
					Diabetes, uncomplicated	18	0.0127932	
					Late effects of cerebrovascular disease	≤10	≤10/1407	
					Pulmonary circulatory disorder	15	0.010661	
					Valvular heart disease	17	0.0120824	
					Peripheral and visceral atherosclerosis	≤10	≤10/1407	
					Any lipid abnormalities	≤10	≤10/1407	
					Lipodystrophy	≤10	≤10/1407	
					Personal history of TIA and cerebral infarction without residual deficits	≤10	≤10/1407	
					Arrythmia	47	0.0334044	
					Valvular heart disease	17	0.0120824	
Adulthood / JDM								
Boros et al. (2022), UK[100]	F:68	Mean 20.6 yrs ±3.9 (SD)	Mean 9.2 yrs ± 4.3 (SD)	Mean 12.4 yrs ± 5.0		84		White 69
					Feel shorter than peers	20 /80	0.25	

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							Feel weight was affected by myositis	25/80	0.3125	
							Puberty affected by myositis	16/80	0.2	
							Calcinosis	18/82	0.225	
							Lipodystrophy	6/82	0.0731707	
							Pulmonary hypertension	2/83	0.0240964	
Mathiesen et al. (2014), Denmark[101]	F: 37	Mean 21.7 yrs (SD 11.1) (Range 4.2-45.9) (Non-restrictive lung), 14.2 (4.4) (10.0-19.7) (restrictive lung)	Mean 7.7 yrs (SD 4.7) (Range 1.4-17.8) (non-restrictive lung), 7.7 (4.2) (1.5-10.5) (restrictive lung)		Mean 4.2 yrs (SD 2.9) (Range 0.610.6) (non-restrictive lung), 5.3 (2.5) (2.6-8.5) (restrictive lung)	Mean 14.3 yrs (SD 10.7) (Range 1.1-35.1) (non-restrictive lung), 6.5 (3.0) (2.6-9.2) (restrictive lung)		51		
							Restrictive interstitial lung disease	4	0.0784314	12.06636501
							Obstructive lung function pattern	7	0.1372549	9.598244892
Tsaltskan et al. (2020), USA[102]	F: 39	Median 24.0 yrs (IQR 19.0-28.6) yrs		Median 11.0 (IQR 8.5-15.7)	median 11.5 yrs (IQR 4.5-18.9) yrs			49		Caucasian 59.2%, African American 11.1% , Hispanic 17.7%, Other 12.0%
							Cutaneous Damage:			
							Cutaneous scarring or atrophy	28	0.5714286	
							Calcinosis	27	0.5510204	
							Lipodystrophy	16	0.3265306	
							Alopecia	8	0.1632653	
							Poikiloderma	6	0.122449	

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Skeletal Damage:		
Joint Contracture	31	0.6326531
Deforming Arthropathy	8	0.1632653
Osteoporosis with fracture	4	0.0816327
Avascular Necrosis	2	0.0408163
Endocrine Damage:		
Irregular Menses	10/40	0.25
Hirsutism	7/40	0.175
Diabetes	6	0.122449
Amenorrhea	4/40	0.1
Infertility	1	0.0204082
Pulmonary Damage:		
Impaired lung function	7	0.1428571
Dysphonia	4	0.0816327
Pulmonary Fibrosis	1	0.0204082
Ocular Damage:		
Cataract	4	0.0816327
Vision Loss	3	0.0612245
Gastrointestinal Damage:		
Dysmotility	5	0.1020408
Dysphagia	3	0.0612245
Infection:		
Chronic Infection	5	0.1020408
Multiple Infections	4	0.0816327
Cardiovascular Damage:		
Hypertension	4	0.0816327
Peripheral Vascular Damage:		
Tissue or pulp loss	4	0.0816327

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					Digit or limb loss	1	0.0204082
					Malignancy	1	0.0204082
Mathiesen et al. (2012), Denmark[103]	F: 38	Mean 21.5 yrs SD 11.1 Range 4.2-45.9)	Mean 4.3 yrs SD 2.9 range (0.6-10.6)	Mean 13.9 SD 10.5 (range 2.1–35.1) yrs		53	
					Deforming arthropathy	1	0.0188679	1.357404642
					Pulmonary fibrosis	3	0.0566038	4.072213927
					Growth failure	6/49	0.122449	8.809279107
					Diabetes	1	0.0188679	1.357404642
					Scarring or atrophy	21	0.3962264	28.50549749
					Calcinosis	11	0.2075472	14.93145107
					Lipodystrophy	7	0.1320755	9.501832496
					Tissue or pulp loss	1	0.0188679	1.357404642
					Alopecia, poikiloderma	0	0	0
					Joint contractures	11	0.2075472	14.93145107
					Osteoporosis/osteopenia estimated by DXA	5	0.0943396	6.787023212
					Osteoporotic fractures	2	0.0377358	2.714809285
					Delay in secondary sexual characteristics	3/14	0.2142857	15.41623844
					Hypertriglyceridemia	9	0.1698113	12.21664178
					Hirsutism, irregular menses, amenorrhoea	0	0	0
					Paraesthesia	2	0.0377358	2.714809285
					Arthritis	3	0.0566038	4.072213927
					Impaired pulmonary function test	4	0.0754717	5.429618569
					Pulmonary hypertension	0	0	0
					Severe abdominal pain	3	0.0566038	4.072213927

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							Dysphagia, gastrointestinal dysfunction	1	0.0188679	1.357404642	
							Gastrointestinal surgery	1	0.0188679	1.357404642	
							Hypertension, cardiomyopathy, angina, myocardial infarction	0	0	0	
							Tissue or pulp loss, claudication	0	0	0	
							Cataract, visual loss	0	0	0	
							Chronic infection, multiple infections	0	0	0	
							Malignancy	0	0	0	
Sanner et al. (2010), Norway[104]	F: 36	Median 21.5 (range 6.7–55.4) yrs	Mean 8.3 ±4.5 yrs	Mean 9.1 ± 4.5 yrs	minimum 24 months	Mean 16.9 ± 10.6		59			European 57
							Calcinosis	14	0.2372881	14.04071808	

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